APPENDIX C
Image Credits

Throughout this book we have made use of publicly available datasets and images available from web services; these were listed in Appendix B. The contributions of the researchers behind these datasets are greatly appreciated.

Some of the reoccurring example images are the author's own. You are free to use these images under a Creative Commons Attribution 3.0 (CC BY 3.0) license (*http://creativecommons.org/licenses/by/3.0/*), for example by citing this book.

These images are:

- The Empire State building image used in almost every example throughout the book.
- The low contrast image in Figure 1-7.
- The feature matching examples used in Figures 2-2, 2-5, 2-6, and 2-7.
- The Fisherman's Wharf sign used in Figures 9-6, 10-1, and 10-2.
- The little boy on top of a hill used in Figures 6-4, 9-6.
- The book image for calibration used in Figures 4-3.
- The two images of the O'Reilly open source book used in Figures 4-4, 4-5, and 4-6.

C.1 Images from Flickr

We used some images from Flickr available with a Creative Commons Attribution 2.0 Generic (CC BY 2.0) license (*http://creativecommons.org/licenses/by/2.0/deed.en*). The contributions from these photographers is greatly appreciated.

The images used from Flickr are (names are the ones used in the examples, not the original filenames):

- *billboard_for_rent.jpg* by @striatic, *http://flickr.com/photos/striatic/21671910/*, used in Figures 3-2.
- *blank_billboard.jpg* by @mediaboytodd, *http://flickr.com/photos/23883605@N06/2317982570/*, used in Figures 3-3.

- *beatles.jpg* by @oddsock, *http://flickr.com/photos/oddsock/82535061/*, used in Figures 3-2, 3-3.
- *turningtorso1.jpg* by @rutgerblom, *http://www.flickr.com/photos/rutgerblom/2873185336/*, used in Figure 3-5.
- *sunset_tree.jpg* by @jpck, *http://www.flickr.com/photos/jpck/3344929385/*, used in Figure 3-5.

C.2 Other Images

- The face images used in Figures 3-6, 3-7, and 3-8 are courtesy of J. K. Keller. The eye and mouth annotations are the author's.
- The Lund University building images used in Figures 3-9, 3-11, and 3-12 are from a dataset used at the Mathematical Imaging Group, Lund University. Photographer was probably Magnus Oskarsson.
- The toy plane 3D model used in Figure 4-6 is from Gilles Tran (Creative Commons License By Attribution).
- The Alcatraz images in Figures 5-7 and 5-8 are courtesy of Carl Olsson.
- The font data set used in Figures 1-8, 6-2, 6-3 6-7, and 6-8 is courtesy of Martin Solli.
- The Sudoku images in Figures 8-6, 8-7, and 8-8 are courtesy of Martin Byröd.

C.3 Illustrations

The epipolar geometry illustration in Figure 5-1 is based on an illustration by Klas Josephson and adapted for this book.

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About the Author

Jan Erik Solem is a Python enthusiast and a computer vision researcher and entrepreneur. He is an applied mathematician and has worked as associate professor, startup CTO, and now also book author. He sometimes writes about computer vision and Python on his blog *www.janeriksolem.net*. He has used Python for computer vision in teaching, research, and industrial applications for many years. He currently lives in San Francisco.

Colophon

The animal on the cover of *Programming Computer Vision with Python* is a bullhead.

Often referred to as "bullhead catfish," members of the genus *Ameiurus* come in three common types: the black bullhead (*Ameiurus melas*), the yellow bullhead (*Ameiurus natalis*), and the brown bullhead (*Ameiurus nebulosus*). These stubborn fish prefer aging, warm-water lakes and are typically found east of the North American continental divide.

Bullheads are known for their obstinacy and tenacity (in fact, people possessing these traits are often characterized as "bullheaded"), and these characteristics have helped them outlive many other species of fish. They can tolerate brackish water and low oxygen and high carbon dioxide levels, making them more resistant to pollutants than most other fish and therefore ideal for lab and medical experiments.

Bullheads are bottom-feeders, prowling in schools at night, on the hunt for clams, insects, leeches, small fish, crayfish, and algae. While doing so, they tend to stir up the bottom of the body of water and destroy aquatic vegetation, which eliminates cover for other species. This, along with their high reproductive rate and subsequent overpopulation, can make them somewhat of a curse for fisheries. For this reason, restrictions are rarely placed on bullhead fishing.

All three types of bullhead are scaleless and average 8 to 10 inches in length. The whisker-like barbells, or feelers, at the corners of their mouth and in a line on the lower chin give them a catfish-like appearance, but they're differentiated partly by the sharp spines at the base of their dorsal and pectoral fins. Like catfish, though, their sense of smell is more developed than that of canines.

The cover image is from Wood's *Animate Creation*. The cover font is Adobe ITC Garamond. The text font is Linotype Birka; the heading font is Adobe Myriad Condensed; and the code font is LucasFont's TheSansMonoCondensed.